Lierda L-LRMAM36-FANN4 Module Communication Distance Test Report

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Document Revision History

Document Version	Change date	Revised by	Reviewer	Change content
Rev1.0	26-06-12	CJH、JHB	LXQ	Initial version

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Applicable Module Selection

Serial number	Module Model	Supported frequency bands	Dimensions (mm)	Module Introduction
1	L-LRMAM36-FANN4	863~930MHz(LoRa) 2400~2500MHz(WiFi&BLE)	20*20*2.5	No firmware/No antenna

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Contents

Legal Disclaimer 1
Document Revision History 1
Contents 2
1 Introduction 3
2 Laboratory performance testing 3
3 Antenna parameters 4
4 Communication distance test 5
4.1 Test environment 5
4.2 Test equipment 6
4.2.1 BLE range testing equipment 6
4.2.2 WiFi range testing equipment 6
4.2.3 LoRa & FLRC Range Testing Equipment 6
4.3 Test procedure 6
4.3.1 BLE Module Range Testing 6
4.3.2 WiFi range test 7
4.3.3 LoRa Range Test 7
4.3.4 FLRC Long Range Test 8
5 Test data and summary 9
6 Module Pull Test Precautions and Instructions 10

1 Introduction

This test report provides range test data for the L-LRMAM36-FANN4 module in an open environment without interference.

2 Laboratory performance testing

Test the module's transmit power, receiving sensitivity, frequency offset and other data under different modes in laboratory conditions.

Table 2-1 Module Sub-G Frequency Band Laboratory Performance Data

Number	Center frequency point (MHz)	Frequency offset (ppm)	Set power	Conducted power (dBm)	LoRa Sensitivity (dBm)	FLRC Sensitivity (dBm)
1#	915	1	22	21.1	-125.5	-98
2#	915	1	22	21.1	-125.3	-98

Table 2-2 Module WiFi & BLE Laboratory Performance Data

Number	Mode	Rate	Frequency offset (ppm)	Transmit power (dBm)	Reception sensitivity (dBm)
3#	WiFi	11b 1M	0.16	19.73	-96
3#	BLE	1M	0.16	0.14	-96

Note:

Remarks

- Test conditions: +25℃;
- LoRa parameter configuration: SF=7, BW=125KHz, CR=4/5;
- FLRC parameter configuration: 2.6Mbps, CR3/4, BT0.5;
- The purpose of this section is to confirm that the test module's Lora, FLRC, WiFi, and BLE performance is normal.

3 Antenna parameters

This test uses an external antenna for Sub-G, and PCB antennas for WiFi & BLE.

The antenna information used for Sub-G communication testing is as follows:

Antenna Model: SLEingA100140201-F16	Antenna gain: 4dBi
Antenna S11 diagram:	Antenna image: 

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4 Communication distance test

4.1 Test environment

The testing environment should be an outdoor open area.



4.2 Test equipment

4.2.1 BLE range testing equipment

Host: iPhone17

Slave: L-LRMAM36-FANN4 Module + AM36 Series Module Development Board 02

4.2.2 WiFi range testing equipment

AP：H3C Magic NX54+PC

STA: L-LRMAM36-FANN4 Module + AM36 Series Module Development Board 02

4.2.3 LoRa & FLRC Range Testing Equipment

- L-LRMAM36-FANN4 Module + AM36 Series Module Development Board 02
- LoRa Antenna SLEingA100140201-F16
- Laptop

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4.3 Test procedure

4.3.1 BLE Module Range Testing

Test preparation:

1. Flash a development board with slave firmware and place it at a fixed height (approximately 1.5m);
2. Download the nRF Connect app on your phone.

Test steps:

1. Set the slave's BLE module transmission power to 0dBm;
2. Continuously scan for Bluetooth devices using a mobile phone, connect to the slave device after scanning it, and record the locations where the device can be successfully scanned and connected to.

3. The slave gradually moves away from the master BLE module, while the master continuously attempts to scan and connect;

4. Record the critical distance at which the signal fails to connect to the device;

5. In locations where you cannot connect to the device, try moving slightly closer to confirm the exact limit of the signal;

6. Repeat the test several times to ensure the accuracy and consistency of the results.

4.3.2 WiFi range test

Test preparation:

1. Flash a development board with WiFi firmware and place it at a fixed height (approximately 1.5m);

2. Install a ping testing tool on the PC, connect to the router via Ethernet cable, and place the router at a fixed height (approximately 1.5m).

Test steps:

1. Connect the module to the router through commands;

2. Record the STA IP address assigned by the router to the module;

3. Use packet loss testing software on the PC;

4. Record the critical distances at which the packet loss rate for 100 packets is approximately 10% and 30% respectively.

5. Repeat the test several times to ensure the accuracy and consistency of the results.

4.3.3 LoRa Range Test

Test preparation:

Burn the LoRa+FLRC test firmware onto two development boards and place them at a fixed height (approximately 1.5m).

Test steps:

1. Set both development boards to LoRa mode, SF=7, BW=125KHz, CR=4/5, with the same frequency point;

2. One development board enters transmission mode, while the other enters reception mode.

3. Two development boards gradually move apart while the transmitter continuously sends packets;

4. Record the critical distance at which the packet loss rate is around 10% when the transmitter sends 100 packets;

5. Repeat the test several times to ensure the accuracy and consistency of the results;

4.3.4 FLRC Long Range Test

Test preparation:

Flash two development boards with LoRa+FLRC test firmware and place them at a fixed height (approximately 1.5m).

Test steps:

1. Set both development boards to FLRC mode, rate 2.6Mbps, CR3/4, BT0.5, with the same frequency point;

2. One development board enters transmission mode, and the other enters reception mode;

3. Two development boards gradually move apart while the transmitter continuously sends packets;

4. Record the critical distance at which the packet loss rate is approximately 10% when the transmitter sends 50 packets;

5. Repeat the test several times to ensure the accuracy and consistency of the results;

6. Set both development boards to FLRC mode, rate 1Mbps, repeat operations 2~5.

5 Test data and summary

Communication Protocol	Test conditions	Installation form	Installation height (cm)	Communication distance (m)
BLE	0dBm，1M，30ms connection interval, 5s connection timeout	Fixed bracket	150	118
WiFi	10% packet loss @ 100 packets	Fixed bracket	150	165
	30% packet loss @ 100 packets	Fixed bracket	150	205
LoRa	10% packet loss, @100 packets SF=7，BW=125kHz，CR=4/5	Fixed bracket	150	2520
FLRC	10% packet loss, @50 packets 2.6Mbps，CR3/4，BT0.5	Fixed bracket	150	515
	10% packet loss, @50 packets 1Mbps，CR3/4，BT0.5	Fixed bracket	150	610

6 Distance Test Precautions and Instructions

(1) During the actual range testing of the module, the current environment has a significant impact on the actual range distance. Factors such as terrain, obstructions, traffic volume, and interference from other airborne devices can all greatly affect the test distance. If the distance is found to be very short during actual communication testing, you can try modifying the communication frequency point to avoid interference from adjacent frequency points of other devices.

(2) During the ranging process, blind spots may exist in certain sections due to terrain or building obstructions. When ranging, you can try to continue moving forward to bypass the blind spots.

(3) The placement position of the communication module, its height from the ground, and antenna direction all affect the distance. When testing range, try to simulate the actual product usage conditions for communication testing. There should be no obstructions near the antennas of the receiving and transmitting modules. Buildings and other obstructions in the transmission path of the module communication will also affect the communication distance.

(4) Since the antennas used generally have certain directivity, when the packet loss rate first exceeds the expected standard, it is necessary to adjust the antenna directions at both the transmitting and receiving ends to determine whether the limit communication distance that meets the packet loss rate has been reached.

(5) To better analyze the range test data, it is necessary to simultaneously record the signal quality at the corresponding communication distance, such as signal RSSI, channel RSSI, and SNR.

(6) If at the extreme communication distance, the RSSI value at the receiving end is still very high with a large gap from the target RSSI, and if conditions permit reading that the SNR is approaching the limit, it can be preliminarily determined that there is interference in the environment or a physical layer parameter configuration problem, and the measured

distance at this time is not the true extreme distance;

(7) If there is a product with the same chip platform that has previously been tested for normal range in pull range testing, a set should be prepared simultaneously for comparison testing.

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